# Where does weight-space programming fix failures?

NVIDIA Cosmos3 Edge Policy (world-action model) and NVIDIA GR00T N1.7 DROID (VLA on a Cosmos-Reason2-2B backbone), with OpenPI π0.5 as the control, on the same RoboLab DROID tasks Cube then banana, Eyeglasses, Cube left of bowl, Yellow on red, Mustard in left bin, Three-cube tower, Flat box, Mustard right of raisin box and a retention task Cube in bowl.

**Method.** When a clearer wording of the same goal makes a robot succeed, copy that wording's internal effect into the weights: one least-squares calculation per layer, no training, and the robot still gets the original instruction.

**π0.5, the control (earlier records).** The edit fixed its two wording failures: Cube then banana 27 → 106 of 128 unseen starts and Mustard right of raisin box 0 → 27 of 64; a random edit of the same size did nothing (9 vs 13 of 64); no edit helped the tasks that fail for other reasons.

**NVIDIA models. The fix:** On NVIDIA's Cosmos3 Edge, a weight edit fixed a task the robot failed only because of how its instruction was worded: 0 → 56 successes from 64 unseen starts. A random edit of the same size, a placebo: 0. Rewording the instruction itself, from the same starts: 56. **Where it works:** Only when written into the part of the model that reads the instruction. **The cost:** It breaks the two other tasks whose instructions say "left": Cube left of bowl 13 → 0 and Mustard in left bin 9 → 0 of 32 unseen starts.



Cosmos3 Edge Policy, Mustard right of raisin box: the wording gap; the clip plays on the live page.

## What weight-space programming did on each task, on unseen starts unless marked practice

| | **Cosmos3 Edge Policy**<br>world-action model | **GR00T N1.7**<br>reasoning-backbone VLA | **π0.5**<br>OpenPI VLA, control |
|---|---|---|---|
| Cube then banana | Mustard right of raisin box edit: 18 → 18 of 32<br>9 / 9; p = 1 | 9 → 0 of 32 practice starts<br>0 / 9, p = 0.0039 | 2 → 40 of 64<br>38 / 0, p = $7.3 \times 10^{-12}$; earlier record: fixed: 27 → 106 of 128 |
| Eyeglasses | Mustard right of raisin box edit: 4 → 1 of 16<br>0 / 3; p = 0.25 | unedited 0 of 32<br>not edited | Cube then banana edit: 12 → 21 of 32<br>12 / 3; p = 0.035; earlier record: Cube then banana edit: 22 → 44 of 64 |
| Cube left of bowl | Mustard right of raisin box edit: 13 → 0 of 32<br>0 / 13; p = $2.4 \times 10^{-4}$ | unedited 1 of 32<br>not edited | Cube then banana edit: 1 → 6 of 32<br>6 / 1; p = 0.12; earlier record: Cube then banana edit: 8 → 36 of 64 |
| Yellow on red | Mustard right of raisin box edit: 24 → 22 of 32<br>4 / 6; p = 0.75 | unedited 0 of 16<br>not edited | no gain earlier slots<br>lookalike perception; no clear wording |
| Mustard in left bin | Mustard right of raisin box edit: 9 → 0 of 32<br>0 / 9; p = 0.0039 | unedited 0 of 32<br>not edited | no gain earlier slots<br>own edit 8 → 11 of 64; the grasp slips |
| Three-cube tower | Mustard right of raisin box edit: 1 → 1 of 32<br>1 / 1; p = 1 | unedited 0 of 16<br>not edited | no gain earlier slots<br>own edit 35 → 39 of 64 |
| Flat box | Mustard right of raisin box edit: 0 → 0 of 16<br>0 / 0; p = 1 | unedited 0 of 32<br>not edited | no gain earlier slots<br>a missing skill; 0 of 64 unedited |
| Mustard right of raisin box | 0 → 56 of 64<br>56 / 0, p = $2.8 \times 10^{-17}$ | unedited 0 of 32<br>not edited | 0 → 16 of 32<br>16 / 0, p = $3.1 \times 10^{-5}$; earlier record: fixed: 0 → 27 of 64 |
| Cube in bowl | 24 → 28 of 32 | unedited 28 of 32<br>not edited | 31 → 31 of 32<br>earlier record: kept: 32 → 32 of 32 |
| Rendering: performance → default preset | — | 5 → 16 of 16<br>Banana onto plate | *running* |
| Random control, same size | control 0, unedited 0, edit 56 of 64<br>Mustard right of raisin box | control 11, unedited 9, edit 0 of 32 practice starts<br>Cube then banana | control 2, unedited 2, edit 40 of 64<br>Cube then banana |
| Where to edit | text pathway 13, video-action (generation) pathway 0 of 16<br>Mustard right of raisin box, 16 practice starts; unedited 0 | Cosmos-Reason2 backbone 0, diffusion action head 0 of 32<br>Cube then banana, 32 practice starts; unedited 9 | VLM 55, action expert 20, head 0 of 64 earlier slots<br>Cube then banana; unedited 5; only the VLM edit transfers |

| | **Cosmos3 Edge** | **GR00T N1.7** | **π0.5** |
|---|---|---|---|
| Cube then banana | 21/32 · 66% | 9/32 · 28% | 3/16 · 19% |
| Eyeglasses | 2/32 · 6% | 0/32 · 0% | 7/16 · 44% |
| Cube left of bowl | 13/32 · 41% | 1/32 · 3% | 0/16 · 0% |
| Yellow on red | 22/32 · 69% | 0/16 · 0% | 9/16 · 56% |
| Mustard in left bin | 11/32 · 34% | 0/32 · 0% | 3/16 · 19% |
| Three-cube tower | 2/32 · 6% | 0/16 · 0% | 6/16 · 38% |
| Flat box | 0/32 · 0% | 0/32 · 0% | 0/16 · 0% |
| Mustard right of raisin box | 0/32 · 0% | 0/32 · 0% | 0/16 · 0% |
| Cube in bowl | 21/32 · 66% | 28/32 · 88% | 16/16 · 100% |

Each cell: the starts the unedited robot completed, of the starts tried, and that share; stronger shading is a higher share.

| | **Cosmos3 Edge** | **GR00T N1.7** | **π0.5** |
|---|---|---|---|
| Cube then banana | ±0 / 32 | **−9 / 32*** | **+38 / 64*** |
| Eyeglasses | −3 / 16 | not run | **+9 / 32*** |
| Cube left of bowl | **−13 / 32*** | not run | +5 / 32 |
| Yellow on red | −2 / 32 | not run | +3 / 32 |
| Mustard in left bin | **−9 / 32*** | not run | −1 / 32 |
| Three-cube tower | ±0 / 32 | not run | +5 / 32 |
| Flat box | ±0 / 16 | not run | ±0 / 32 |
| Mustard right of raisin box | **+56 / 64*** | not run | not run |
| Cube in bowl | +4 / 32 | not run | ±0 / 32 |

gain, $p < 0.05$ loss, $p < 0.05$ no detectable change

Cell: change in starts completed / starts tried. * $p < 0.05$; outlined: the edit's own task

Each model's main edit, unchanged, on every suite task: the change in starts completed, edited minus unedited, from the same starts. Teal: a gain; near-black (near-white in dark mode): a loss (exact McNemar $p < 0.05$); grey: no detectable change; the outlined cell is the task the edit was built for.

Live page with every table, figure and record: ali-adeeb.com/weightsmith/cosmos · Paper: ali-adeeb.com/weightsmith/paper.pdf · Code and records: github.com/adeeb10abbas/weightsmith

Draft for discussion. Simulation only (RoboLab, Isaac Sim), one released checkpoint per model, shared slots 30000–30031 and 96000–96063 with identical initial scenes; π0.5 cells are earlier records on their own slots until its shared-slot runs land. Exact McNemar tests on paired starts.